

Value Object Boundary Module (VOBM)

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Architecture Ecosystem: Structured Reality Standards™

Architecture Family: VFM® — Value Flow Mechanisms Architecture

Parent Standard: Value Identification Standard (VIS)

Module: VOBM — Value Object Boundary Module

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Governed Space: Value Object Boundary

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Canonical Language: English

Canonical Definition

Value Object Boundary Module (VOBM) is the second Core Module of the Value Identification Standard (VIS) within VFM® — Value Flow Mechanisms Architecture.

VOBM governs the methodological boundary of an identified Value Object by determining what belongs inside the Value Object for the relevant value lifecycle, what remains outside it, which components, relationships, capabilities, conditions, dependencies, or interfaces materially constitute the object, and where the object must be separated from its surrounding environment.

The Module transforms an identified Value Object into a bounded Value Object capable of supporting consistent downstream value governance.

VOBM does not determine whether value exists, where value originates, whether value has emerged, how much value exists, or how value changes.

Its foundational task is narrower:

to establish where the Value Object begins and ends before the value associated with that object enters the wider value lifecycle.

Core Question

Where does the identified Value Object begin and end for the value lifecycle being governed?

Architectural Position

VOBM is the second Core Module of the Value Identification Standard.

It follows:

VOIM — Value Object Identification Module

and precedes the subsequent VIS modules governing differentiation, context, and the remaining conditions necessary to establish the Value Object as a governed entry point into the value lifecycle.

VOIM establishes:

What is the Value Object?

VOBM establishes:

What belongs to that Value Object, and where does it end?

The sequence is therefore:

Value Object Identification → Value Object Boundary → Value Differentiation → Value Context → Governed Value Identification

Internal Flow Position

Identified Value Object → Boundary Purpose Definition → Constituent Identification → Inclusion Assessment → Exclusion Assessment → Dependency Assessment → Interface Identification → Composite / Component Boundary Assessment → Temporal Boundary Assessment → Boundary Determination → Boundary State Assignment → Bounded Value Object

Operational Space

VOBM governs the space between:

an identified Value Object

and

a sufficiently bounded Value Object capable of entering further value governance.

This operational space includes:

Value Object boundary purpose; object inclusion; object exclusion; constituent identification; component boundaries; composite-object boundaries; relational boundaries; capability boundaries; process boundaries; service boundaries; digital boundaries; physical boundaries; functional boundaries; dependency boundaries; interface boundaries; contextual boundaries where necessary for object separation; temporal boundaries; configuration boundaries; boundary uncertainty; boundary change; boundary state; boundary traceability.

Why Value Boundaries Matter

Value cannot be governed consistently where the object carrying, enabling, producing, receiving, or relating to that value changes silently during analysis.

Consider an industrial facility.

One analysis may treat the Value Object as:

the land.

Another may treat it as:

land + buildings.

Another:

land + buildings + machinery.

Another:

the complete operational production capability.

Another:

the business operating from that facility.

All may appear to discuss the same asset.

They are not necessarily discussing the same Value Object.

Their boundaries differ.

VOBM exists to make that difference explicit.

Value Object Boundary

A Value Object Boundary is:

the governed methodological separation defining which elements, components, relationships, capabilities, conditions, and interfaces constitute the Value Object for the relevant value lifecycle and which remain outside it.

The boundary does not need to correspond to:

a physical perimeter; legal ownership; accounting consolidation; organizational structure; technical architecture; contractual structure; regulatory jurisdiction.

Those boundaries may provide relevant inputs.

They do not automatically determine the VFM® Value Object Boundary.

Physical Boundary ≠ Value Object Boundary

A physical asset may have a clear physical perimeter while the relevant Value Object extends beyond it.

For example, a factory building has a physical boundary.

But the Value Object:

Operational Production Capability

may depend upon:

machinery; energy access; workforce capability; software; process knowledge; supply access; logistics; licenses; external infrastructure.

The physical factory perimeter therefore does not automatically define the boundary of the relevant Value Object.

Conversely, the relevant Value Object may concern only one machine located inside the factory.

Therefore:

Physical Boundary ≠ Value Object Boundary

Legal Boundary ≠ Value Object Boundary

Legal ownership may identify who owns an object or right.

It does not automatically determine what constitutes the Value Object for a specific value lifecycle.

A company may legally own:

land;

machinery; patents; software; inventory; contractual rights.

But a particular Value Object may consist of only one of those elements or a governed combination of several.

Therefore:

Ownership Boundary ≠ Value Object Boundary

Ownership remains governed outside VOBM where relevant.

Accounting Boundary ≠ Value Object Boundary

Accounting systems may group objects according to accounting rules.

VOBM does not assume that an accounting asset corresponds exactly to a Value Object.

One accounting asset may contain multiple value-relevant objects.

Multiple accounting assets may together form one Value Object.

Some Value Objects may not be represented as conventional balance-sheet assets at all.

Examples may include:

dormant capability; network position; operational access; organizational knowledge; technological potential; unused capacity; relational capability; data-derived capability.

Therefore:

Accounting Asset Boundary ≠ Value Object Boundary

Boundary Purpose

Every boundary determination should identify why the Value Object is being bounded.

The purpose may relate to:

value lifecycle analysis; value origin examination; value emergence; value state determination; value change analysis; value degradation; value nullification; value restoration; value extinction; comparison between Value Objects; organizational decision support; policy analysis; economic analysis; asset restructuring; other governed value purposes.

Boundary purpose prevents the same object from being bounded differently without explanation.

Constituent Identification

VOBM identifies the constituents potentially forming part of the Value Object.

Constituents may include:

physical components; digital components; information objects; capabilities; rights; processes; activities; relationships; resources; access conditions; infrastructure dependencies; contractual elements; operational elements; environmental

elements; other materially relevant constituents.

Constituent identification does not automatically mean inclusion.

Each constituent must be assessed against the boundary purpose.

Inclusion Principle

An element should be included within the Value Object Boundary where its presence is materially constitutive of the object being governed.

The relevant question is:

Would removing this element materially change what the identified Value Object is?

Where the answer is yes, inclusion may be justified.

Where the element merely affects the Value Object externally, it may instead remain an external dependency or contextual condition.

This distinction is essential.

Exclusion Principle

VOBM requires explicit exclusion of elements that are adjacent, influential, related, or economically relevant but do not constitute the Value Object itself.

Examples may include:

market conditions; taxation; financing conditions; regulation; external demand; external infrastructure; customer behavior; supplier conditions; macroeconomic conditions.

These factors may strongly affect value.

That does not automatically make them part of the Value Object.

Therefore:

Influence on Value $\neq$ Membership in Value Object

Constituent $\neq$ Dependency

VOBM distinguishes:

Constituent

from

Dependency.

A constituent forms part of the Value Object.

A dependency exists outside the object but materially affects its ability to function, exist, or produce relevant value.

For example:

A data center may include servers as constituents.

Electric-grid access may remain an external dependency.

But where the Value Object is defined as:

Integrated Data-Center Operational Capability

the relevant energy infrastructure relationship may need to be represented differently.

Boundary determination therefore depends upon the precise Value Object identified by VOIM.

Dependency Boundary

External dependencies must remain identifiable even when they are excluded from the Value Object itself.

A bounded Value Object may therefore be represented as:

Internal Constituents → Object Boundary → External Dependencies

This enables later VFM® analysis to distinguish:

change to the Value Object

from

change in an external condition affecting the value of the Value Object.

Interface Boundary

Where the Value Object interacts with external systems, VOBM identifies the relevant interfaces.

Interfaces may include:

physical interfaces; digital interfaces; contractual interfaces;

resource interfaces; information interfaces; service interfaces; infrastructure interfaces; operational interfaces; market interfaces.

The interface defines where interaction occurs.

It does not automatically move the external system inside the Value Object Boundary.

Composite Value Object Boundary

A composite Value Object may contain multiple independently identifiable components.

For example:

Integrated Manufacturing Capability

may include:

Facility + Machinery + Control Software + Process Capability + Defined Operational Integration

VOBM must determine whether the governed lifecycle concerns:

each component separately

or

the integrated composite object.

This distinction is necessary because:

Value of Composite Object $\neq$ Sum of Isolated Component Values

A composite object may possess value that emerges only through integration.

Likewise, loss of one critical constituent may disproportionately degrade the value of the complete object.

Component Value Object Boundary

A component inside a composite Value Object may itself have an independent value lifecycle.

VOBM therefore supports nested boundaries:

Composite Value Object A → Component Value Object A.1 → Component Value Object A.2 → Component Value Object A.3

Each boundary must remain distinguishable.

This enables VFM® to govern:

component value

and

composite value

without collapsing them into one undifferentiated value object.

Relational Value Object Boundary

Some Value Objects exist primarily through relationships.

Examples may include:

distribution relationships; network participation; access arrangements; partnerships; contractual opportunities; platform relationships; supply relationships.

In such cases, VOBM must determine what constitutes the relationship-based Value Object without treating the complete identities, organizations, or assets of all participating parties as part of that Value Object.

Therefore:

Relationship Participation ≠ Full Object Inclusion

Capability Boundary

Capabilities often create difficult boundary questions.

A production capability may depend on:

equipment; people; knowledge; software; infrastructure; procedures; permissions; supply availability.

VOBM determines which of these constitute the identified capability Value Object and which remain enabling dependencies or contextual conditions.

This prevents capability value from being confused with the value of every object contributing to that capability.

Process Boundary

Where a process is the Value Object, the boundary may be defined through:

process entry; process activities; transformation stages; required resources; process exit; interfaces; dependencies.

VOBM does not govern operational execution of the process.

It establishes what process is included within the Value Object being examined.

Temporal Boundary

A Value Object Boundary may change over time.

For example:

Factory Capability — 2024

may include one production line.

After expansion:

Factory Capability — 2026

may include three lines, automation infrastructure, and a new digital control system.

VOBM therefore requires material boundary changes to remain visible.

A historical value determination must not silently inherit a later object boundary.

Configuration Boundary

Where configuration materially determines the Value Object, VOBM records the configuration associated with the boundary.

Examples include:

industrial system configuration; software stack; AI system configuration; infrastructure topology; portfolio composition; organizational composition; service bundle; production configuration.

A material configuration change may require a new or revised Value Object Boundary.

Dynamic Boundaries

Some Value Objects have boundaries that change during their lifecycle.

Examples may include:

evolving digital platforms; portfolios; supply networks; project ecosystems; integrated capabilities; data environments.

VOBM therefore permits:

Dynamic Value Object Boundaries

provided that the basis, time, conditions, and material changes remain traceable.

Dynamic does not mean undefined.

Boundary Uncertainty

Not every Value Object can initially be bounded with complete certainty.

VOBM therefore permits boundary uncertainty to be explicitly represented.

Uncertainty may concern:

component inclusion;

dependency classification; relationship extent; temporal applicability; configuration; interface definition; composite structure.

Uncertainty must not be hidden through arbitrary precision.

Where material uncertainty remains, the boundary may receive a conditional state.

Boundary States

VOBM may assign a governed boundary state.

1. Unbounded

The Value Object has been identified but its relevant boundary has not yet been established.

2. Candidate Boundary

A proposed boundary exists but requires further assessment.

3. Conditionally Bounded

The boundary is sufficiently established only under defined assumptions, contexts, configurations, or temporal conditions.

4. Bounded

The Value Object Boundary is sufficiently established for downstream value governance.

5. Boundary Under Review

A material change or unresolved issue requires reassessment.

6. Boundary Superseded

A previously governed boundary has been replaced by a newer boundary while remaining preserved for historical traceability.

Boundary states must not be confused with Value States.

Boundary State ≠ Value State

A Value Object may be:

Bounded

while its value is:

unknown, emerging, positive, developing, degrading, zero, restored, or extinct.

VOBM governs the object boundary.

Later VFM® standards govern value states and transitions.

Boundary Change

A boundary change occurs when the governed constitution of the Value Object materially changes.

Examples include:

component addition; component removal; merger; separation; integration; configuration change; relationship termination; capability expansion; capability contraction; structural reorganization.

A boundary change may affect value.

But:

Boundary Change $\neq$ Value Change

The boundary change is an event concerning the Value Object.

Later VFM® governance determines what happened to value as a result.

Boundary with VOIM

VOIM answers:

What is the Value Object?

VOBM answers:

Where does that Value Object begin and end?

Therefore:

Identification $\neq$ Boundary Determination

VOIM must provide a sufficiently identified Value Object before VOBM can establish its boundary.

Boundary with Value Differentiation &

Separation

VOBM determines what is inside and outside the Value Object.

The subsequent differentiation function determines whether multiple objects, value relationships, value types, components, or value phenomena must remain analytically separate.

Therefore:

Boundary $\neq$ Differentiation

An object can be correctly bounded while still containing several distinct value relationships requiring separation.

Boundary with Value Context & Condition

VOBM may use context where necessary to establish the object boundary.

The Value Context & Condition function governs the broader circumstances under which the object enters value analysis.

Therefore:

Boundary Context $\neq$ Value Context Governance

Boundary with Value Origin Standard

VOBM does not determine where value comes from.

It establishes the bounded object in relation to which Value Origin can subsequently be examined.

Therefore:

Boundary with Value Emergence Standard

VOBM does not determine whether value has emerged.

A perfectly bounded Value Object may still possess:

no emerged value.

Therefore:

Bounded Object ≠ Existing Value

Boundary with Value Recognition Standard

VOBM does not determine whether value is recognized.

It establishes only the object boundary required for consistent value governance.

Boundary with GOA™

GOA™ governs governance boundaries.

VOBM governs the boundary of the Value Object for the value lifecycle.

Therefore:

Governance Boundary ≠ Value Object Boundary

Boundary with ORA™

ORA™ governs actual Operational Reality and operational boundaries.

VOBM may use operational information when defining a Value Object but does not govern Operational Reality itself.

Therefore:

Operational Boundary ≠ Value Object Boundary

Boundary with OBIDENTITY®

OBIDENTITY® governs origin-bound identity, ownership, authority, permission, responsibility, provenance, transfer, and continuity.

VOBM does not determine ownership boundaries or identity boundaries.

A Value Object may cross multiple ownership boundaries.

Likewise, one owner may hold multiple distinct Value Objects.

Boundary with CLA™

VOBM may identify constituents and boundary forms necessary for object separation.

Formal classification remains governed by CLA™ where required.

Boundary with VALIDOS®

VOBM does not validate the correctness of a boundary claim.

It defines the methodological boundary required by VFM®.

Where formal validation of the boundary or supporting claims is required, VALIDOS® governs that validation process.

Boundary with INTEGROS®

VOBM does not determine boundary integrity.

INTEGROS® may govern whether a defined boundary remains structurally or operationally intact where integrity governance is required.

Boundary with ADIT®

VOBM establishes the Value Object Boundary.

ADIT® may provide governed audit evidence concerning whether the recorded constituents, exclusions, dependencies, interfaces, or boundary changes are supported by evidence.

Boundary Governance ≠ Audit Governance

Minimum Implementation Framework

A minimum VOBM implementation should establish:

1. Identified Value Object

Receive the identified Value Object from VOIM.

2. Boundary Purpose

Define why the object boundary is required.

3. Constituent Candidates

Identify components, relationships, capabilities, processes, resources, or other elements potentially belonging inside the object.

4. Inclusion Assessment

Determine which constituent candidates form part of the Value Object.

5. Exclusion Assessment

Determine which adjacent or influential elements remain outside the object.

6. Dependency Assessment

Identify external dependencies without automatically incorporating them into the object.

7. Interface Identification

Define relevant interfaces between the Value Object and external systems or environments.

8. Composite / Component Assessment

Determine whether nested Value Object boundaries are required.

9. Temporal & Configuration Reference

Establish the time and configuration to which the boundary applies.

10. Boundary State

Assign the governed Value Object Boundary State.

11. Boundary Record

Document the basis, inclusions, exclusions, dependencies, interfaces, uncertainty, and applicable conditions.

Governance Outputs

VOBM may produce:

Value Object Boundary Record Value Object Boundary Statement Value Object Constituent Register Value Object Inclusion Record Value Object Exclusion Record Value Object Dependency Record Value Object Interface Record Composite Value Object Boundary Record Component Value Object Boundary Record Relational Value Object Boundary Record Value Object Temporal Boundary Value Object Configuration Boundary Value Object Boundary Uncertainty Record Value Object Boundary State Value Object Boundary Change Record Bounded Value Object

These outputs provide the governed basis for the next VIS module.

Enterprise Application

VOBM enables organizations to resolve one of the most important hidden problems in value analysis:

different assessments often assign value to differently bounded objects while treating them as though they were the same object.

This can affect:

enterprise valuation; infrastructure; industrial assets; digital assets; intellectual property; data; production capacity; environmental resources; public assets; projects; services; technology platforms; dormant assets; composite systems; capabilities.

For example, two analysts may disagree about the value of a production facility.

One values:

land + building + machinery.

The other values:

the complete operational production capability, including integrated software, process capability, and operational access.

Their numerical disagreement may be secondary.

Their first disagreement is architectural:

they have not bounded the same Value Object.

VOBM makes that difference visible before downstream valuation, accounting, taxation, investment, policy, or economic analysis attempts to use the resulting value information.

Foundational Principles

A Value Object cannot be governed consistently if its boundary is undefined.

Identification establishes what the Value Object is; boundary governance establishes where it ends.

Physical boundary is not automatically Value Object Boundary.

Legal ownership boundary is not automatically Value Object Boundary.

Accounting asset boundary is not automatically Value Object Boundary.

Influence on value does not automatically create membership in the Value Object.

A constituent is not the same as a dependency.

A composite Value Object may possess a value lifecycle distinct from the value lifecycles of its components.

A component may simultaneously be part of a composite Value Object and possess its own independently governed value lifecycle.

Dynamic boundaries must remain governed boundaries.

Boundary change is not automatically value change.

Boundary State is not Value State.

Canonical Closing Statement

Value Object Boundary Module establishes the second methodological control point of the Value Identification Standard by determining where an identified Value Object begins and ends for the relevant value lifecycle. Through governance of constituents, inclusions, exclusions, dependencies, interfaces, composite and component structures, relational and capability boundaries, temporal applicability, configuration, uncertainty, boundary states, and boundary change, VOBM prevents value analysis from silently expanding, contracting, combining, or substituting the object whose value is being governed. By separating Value Object Boundary from physical perimeter, legal ownership, accounting treatment, governance boundary, operational boundary, classification, validation, integrity, and audit, VOBM creates a stable methodological object upon which subsequent value governance can operate. Before value origin, emergence, recognition, state, development, degradation, nullification, restoration, or extinction can be governed consistently, the boundaries of the object to which that value relates must first be explicit.

Modul3

Governed Links

[→ Value Identification Standard \(VIS\)](#)

[→ VFM® Architecture Map](#)

[→ Previous: Value Object Identification Module \(VOIM\)](#)

[→ Next: Value Object Differentiation Module \(VODM\)](#)

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